

ALEJANDRO ITO ARAMENDIA

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EDUCATION

Tohoku University, Japan – *BEng Mechanical and Aerospace Engineering* 2021 – 2025

- GPA: 3.35, Finite Element Method Analysis (Python), Pygame (Python)

King's College London Maths School, UK – *A-Levels* 2019 – 2021

- A*AAA in Maths, Further Maths, Physics and Computer Science (AS)
- Virtual Sprint Business Idea Competition Winner
- Dynamics of Swing Pumping - King Certificate Project

EXPERIENCE

JIGE Internship Business Competition – *Kansai University* Sep – Nov 2024

- Led cross-functional team in market analysis and strategic planning, identifying growth opportunities and addressing supply chain risks for Peel Lab's European expansion.
- Developed data-driven insights and business metrics to prioritize target sectors, aligning expansion goals with market demand and competitive analysis.

Junior Researcher – *Ishii Lab, Tohoku University* Oct 2023 – Oct 2024

- Developed a MATLAB GUI to automate and model complex vector fluid flow, currently being used by researchers to analyse multiple datasets and increase productivity.
- Used regression techniques in MATLAB to analyse the propagation speed of urethra tissue when voiding.

English Teacher – *Lanny's English School, Sendai* Oct 2023 – Present

- Teaching English to Japanese kids and adults, ensuring that every lesson is taught in a clear structured manner, emphasising greatly on communication within the class.

International PBL Workshop 2023 – *National Taipei University of Technology* Aug 2023

- Created an Arduino robot in C++ that could differentiate coloured balls and take them to designated locations accordingly.
- For two weeks, I led the presentations for my team, communicating our progress and goals in a compendious manner to the judges and 100 other participants.

PROJECTS

SRS Japanese Flashcard Website Apr – Sep 2024

- Developed a flashcard system in JavaScript with exclusive functionalities making learning kanji easier for users.
- Utilised Redis, PostgreSQL and bcrypt authentication to safely store user and flashcard data.
- Required knowledge in API Design, User Authentication and Database Handling.

Chess Game Mar – Apr 2024

- Developed a fully functioning chess game in JavaScript, HTML and CSS.
- Functionalities such as displaying legal moves with red dots required complex problem solving skills to navigate through the numerous move patterns that arise during a game.

Medium Writer on Machine Learning and Algorithms (2000+ Likes) Jan 2024 – Present

- **Articles on ML:** Transformers, CNN, DCNN, U-Net, U-Net++, DenseNet, L1 and L2 Regularization
- **Articles on DSA:** Dijkstra, Quick Sort, Hash Tables, Merge Sort, Time Complexity, Bubble Sort

SKILLS

Programming – JavaScript, Node.js, Express, Redis, Authentication, Cookies, HTML/CSS, SQL, Python, MATLAB
Languages – English (Native), Spanish and Japanese (Limited Professional Proficiency)